

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELAGAVI – 590018



A Project Report on
**“PREDICTION OF LUMPY SKIN
DISEASE IN COW’S USING DEEP
MACHINE LEARNING”**

Submitted in partial fulfillment of the requirements for the award of degree

of

BACHELOR OF ENGINEERING

IN

INFORMATION SCIENCE AND ENGINEERING

Submitted by:

DIVYA R [1JT22IS016]

REETHU S [1JT22IS037]

SANJANA C [1JT22IS043]

VANDANA C [1JT22IS060]

Under the Guidance of

Mr. Manjesh B N

**Assistant Professor, Dept. of ISE,
JIT, Bengaluru.**



DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

JYOTHY INSTITUTE OF TECHNOLOGY

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JYOTHY INSTITUTE OF TECHNOLOGY
Off Kanakapura road, Tataguni, Bangalore-560082.

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING



CERTIFICATE

Certified that the project work entitled “**Prediction Of Lumpy Skin Disease In Cow’s Using Deep machine learning**” carried out by **Divya R [1JT22IS016]**, **Reethu S [1JT22IS037]**, **Sanjana C [1JT22IS043]**, **Vandana C [1JT22IS060]** a bonafide students of “**Jyothy Institute of Technology**” in partial fulfillment of the requirements for the award of the degree of “**Bachelor of Engineering**” in **Information Science and Engineering of Visvesvaraya Technological University, Belagavi**, during the year 2025-26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the Departmental library. The Project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said degree.

Signature of the Guide

Mr. Manjesh B N

Assistant Professor

Dept. of ISE, JIT

Signature of the HOD

Dr. Kishore G R

Assistant Professor and Head

Dept. of ISE, JIT

Signature of the Principal

Dr. Gopala Krishna N

Principal

JIT, Bengaluru

Name of the Examiners:

1. _____

2. _____

Signature with Date:

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Regards,

DIVYA R	[1JT22IS016]
REETHU S	[1JT22IS037]
SANJANA C	[1JT22IS043]
VANDANA c	[1JT22IS060]

DECLARATION

We, the students of the final semester of Information Science and Engineering, Jyothy Institute of Technology, Tataguni-560082 declare that the work entitled **“Prediction Of Lumpy Skin Disease In Cow’s Using Deep machine learning”** has been successfully completed under the guidance of Manjesh B N, Asst. Professor, Department of Information Science and Engineering, Jyothy Institute of technology, Tataguni. This dissertation work is submitted to Visvesvaraya Technological University in partial fulfilment of the requirements for the award of Degree of Bachelor of Engineering in Information Science and Engineering during the academic year 2025 – 2026. Further the matter embodied in the project report has not been submitted previously by anybody for the award of any degree or diploma to any university.

Place: Bangalore

Date:

DIVYA R	[1JT22IS016]
REETHU S	[1JT22IS037]
SANJANA C	[1JT22IS043]
VANDANAC	[1JT22IS060]

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CHAPTER 1

INTRODUCTION

INTRODUCTION

Lumpy Skin Disease (LSD) is a contagious viral disease that affects cattle, primarily transmitted by biting insects such as mosquitoes, flies, and ticks. It is characterized by the appearance of large skin nodules distributed across the animal's body. These nodules can be widespread or limited in number and may eventually ulcerate and leave scars. Along with nodules, common clinical symptoms include fever, nasal discharge, swollen superficial lymph nodes, and tearing (lachrymation).

First identified in Egypt, LSD has become endemic in various regions, including Africa, Russia, Egypt, Oman, India, and has more recently spread to Turkey, the Balkans, Georgia, Bangladesh, China, and several Middle Eastern countries. Its rapid geographical spread has raised international concerns. Notably, LSD has not yet been reported in the Western Hemisphere, Australia, or New Zealand.

Transmission occurs mainly through insect bites, but the virus can also spread through direct contact with skin lesions, saliva, nasal discharge, milk, or semen of infected animals. LSD can occur epidemically or sporadically. New outbreaks may arise far from initial infection sites, especially during wet summer seasons and in regions near water bodies or low-lying areas. Although quarantine and movement restrictions are imposed, they are often ineffective due to insect vectors and other possible routes of transmission.

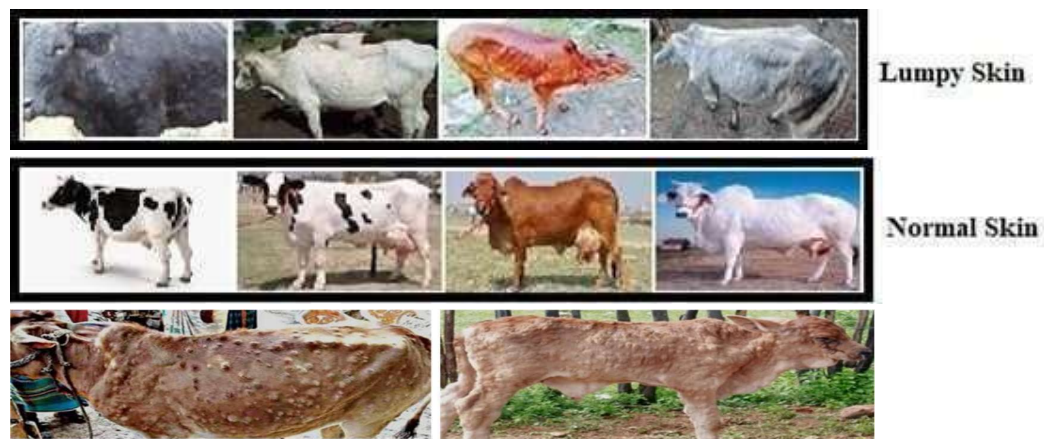


Fig 1.1. Images of Lumpy Skin and Normal Skin in cows.

Experimentally, three species of hard ticks in Africa have been found capable of biologically transmitting the virus. In addition to cattle, African buffalo are suspected reservoir hosts, and other wildlife species might also play a role in maintaining and spreading the disease.

Clinically, diagnosis is based on visible signs like skin nodules, fever, and enlarged lymph nodes. However, confirmation of LSD requires laboratory testing using blood and tissue samples. Currently, there are no commercial diagnostic kits available, and identification largely depends on clinical observation, differential diagnosis, and lab confirmation. There is no specific antiviral treatment available

for LSD. Instead, management focuses on supportive care, such as treating skin lesions with wound care sprays, using antibiotics to prevent secondary bacterial infections and pneumonia, and ensuring good animal husbandry practices to reduce stress and promote healing. In recent years, Machine Learning (ML) has emerged as a powerful tool in veterinary science and environmental data analysis. With the availability of large datasets, ML techniques— particularly deep learning—are increasingly used to identify patterns, conduct spatial-temporal analysis, and create decision-support tools. For instance, ML can be applied to detect and classify cows infected with LSD based on image data, potentially improving the early diagnosis and control of the disease.

1.1 Motivation

Lumpy Skin Disease (LSD) is a highly contagious virus that causes serious health problems in cattle and costs farmers a lot of money, particularly in underdeveloped nations. The majority of traditional disease detection techniques are reactive and time-consuming, which causes treatment to be delayed and disease to spread quickly through insect vectors like flies and mosquitoes. The development of Deep Machine Learning (DML) has made it feasible to use imaging data, historical records, and environmental factors to assess illness trends for early prediction. Farmers and veterinarians may take preventive measures, lower losses, enhance cow health, and improve livestock disease management systems with the help of an accurate and fast prediction system.

1.2 Objectives of the Project

- To facilitate reliable experimentation, an extensive dataset on lumpy skin diseases is assembled.
- For accurate lesion identification, sophisticated segmentation and prediction techniques are chosen.
- Effective training, testing, and automated diagnosis are made possible by an intuitive image processing model.
- A trained model is developed to predict and differentiate lumpy skin disease.
- A WhatsApp-based communication interface enhances accessibility and simplifies disease detection for farmers.
- Early detection significantly reduces disease spread and supports large-scale livestock health monitoring.

1.3 Scope of the Project

- Focuses on predicting Lumpy Skin Disease in cows using Deep Machine Learning techniques.
- Utilizes image processing for accurate lesion detection and disease classification.
- Enables early diagnosis to reduce disease spread and economic losses.
- Integrates a WhatsApp-based interface for farmer-friendly and remote disease monitoring.

1.4 Problem Statement

Diseases in cows are an influential point for human concern. There are some diseases in animals identified in the early phases that can be diagnosed and cured in the early phases of the disease itself. The effect of lumpy skin disease can cause large capital losses in the farm animal industry if it is not taken care of properly. The main reason for this disease is the lumpy skin virus, and this virus is a part of the Poxviridae family.

1.5 Problem Description

Farmers suffer significant financial losses due to Lumpy Skin Disease (LSD), a deadly viral disease that affects cattle. Insects like ticks, flies, and mosquitoes are the primary means of disease transmission. Skin nodules, fever, decreased milk output, infertility, and in extreme situations, death, are all symptoms experienced by infected cows. Because current diagnostic techniques rely on laboratory testing and manual examination by vets, which are time-consuming and difficult to reach in rural regions, early diagnosis of LSD is challenging. The illness frequently spreads quickly and causes widespread epidemics as a result of delayed diagnosis. Timely prevention and control are further hampered by the absence of automated illness detection systems. As a result, an effective and trustworthy early detection technique is desperately needed. To predict LSD early, enhance cow health management, and lower financial losses, a Deep Machine Learning-based method can examine illness patterns and visual signs.

CHAPTER 2

LITERATURE SURVEY

LITERATURE SURVEY

2.1 Tuppurainen, Eeva, (2017), Epidemiology of Lumpy skin disease.

Outbreaks of lumpy skin disease (LSD) typically occur as epidemics separated by several years (Davies, 1991). It is not known where the virus resides between epidemics. Outbreaks are usually seasonal, but they are not restricted to warm and humid seasons with abundance of blood-feeding vectors: in many of the regions affected there is no completely vector-free season, so year-round outbreaks can occur. Morbidity varies between 2 percent and 45 percent; mortality is usually less than 10 percent (Coetzer, 2004). The susceptibility of host animals depends on immune status, age and breed rather than the virulence of the virus. European high-producing cattle breeds are generally more susceptible than indigenous African and Asian breeds (Tageldin et al., 2014); cows at the peak of milk production are usually the most severely affected.

2.2 Shadab, Malik & Dwivedi, Mahavir & S N, Omkar & Javed, Tahir & Bakey, Abdul & Raqib, Mohammad & Chakravarthy, Akshay. (2019). Disease Recognition in Sugarcane Crop Using Deep Learning. 10.13140/RG.2.2.21849.47209.

Recognizing crop diseases remains a major challenge in agriculture, but advancements in visual computing and CNNs have enabled automated disease detection. This study focuses on identifying five sugarcane plant diseases using images collected from fields across Karnataka, India, captured under varying resolutions and lighting conditions. CNN-based models trained on this diverse dataset achieved 93.20% accuracy on the test set and 76.40% accuracy on images from trusted online sources, proving their robustness in real-world conditions. To further localize infected regions, two object detection algorithms—YOLO and Faster-RCNN—were applied, with the best Mean Average Precision (mAP) of 58.13%. The results highlight the potential of deep learning models in reliably identifying crop diseases under uncontrolled field environments, paving the way for effective and scalable automated disease recognition systems in agriculture.

2.3 Sandeep Sharma, Kapil Joshi, Rajesh Singh, Priyanka Sumit Chetwani, Amandeep Kaur, Gaurav Kumar (2024) – Design of an Improved Model for Lumpy Skin Disease Prediction Using Graph Neural Networks and Multiple Modal Data Fusion.

This study introduces a graph neural network (GNN)–based predictive model that integrates diverse data including epidemiological metrics, livestock movements, weather information, and satellite imagery using a multi-head attention mechanism. They first construct spatial-temporal embeddings via graph convolution, then fuse different modalities and apply contrastive (SimCLR) self-supervised pre-training. This holistic architecture achieved 95% overall accuracy, with notable resilience to adversarial scenarios and informative attention weights aiding model interpretability. This approach addresses limitations of traditional uni-source models by leveraging rich, multi-dimensional feature spaces and modern neural methods to provide accurate and robust predictions. The authors highlight the model's applicability in real-world veterinary surveillance systems, offering proactive outbreak detection and targeted control strategies.

2.4 Girma et al. cited within BMC Infectious Diseases (2024) – Prediction of Lumpy Skin Disease Virus Using Customized CBAM-Dense Net Attention Model.

This paper presents an attention-enhanced Dense Net (CBAM-Dense Net) designed to classify LSD images. The model was benchmarked against previous approaches: Girma et al. (2021) with CNN+SVM (96.5% accuracy), a 2022 ANN model (98.6%), and Mobile Net-based architectures (~86–97%). The CBAM-Dense Net approach outperforms these baselines, achieving 99.11% accuracy and 98.99% F1-score in a three-class problem (LSD, monkeypox, chickenpox). By combining convolutional architectures with channel and spatial attention (CBAM), the model precisely highlights key visual features associated with LSD. This contributes not only to classification accuracy but also to enhanced interpretability useful for field deployment and integration with image-based disease surveillance tools.

2.5 Olatayo Olaniyan, Olusogo Julius Adetunji et al. (2023) – Development of a Model for the Prediction of Lumpy Skin Diseases using Machine Learning Techniques.

This paper compares several traditional machine learning methods—including Decision Trees (DT), k-Nearest Neighbors (KNN), Random Forests (RF), and Support Vector Machines (SVM)—on a dataset of LSD outbreaks collected between 2005 and 2021. After applying feature selection (via Featurewiz) and parameter tuning, the researchers implemented both a stacked ensemble of these classifiers and a tuned ANN model. The ensemble achieved 97.7% accuracy, precision 98.4%, F1-score 98.9%, and recall 98.7%, while the optimized ANN surpassed it with 98.9% training accuracy and 98.7% validation accuracy. This work demonstrates that deep learning approaches, when well-tuned, can outperform sophisticated ensembles built from traditional ML models for predicting LSD outbreak risk. Their findings underscore the value of combining feature selection techniques with both ensemble and deep neural methods to enhance predictive performance in epidemiological surveillance.

2.6 Ehsanallah Afshari Safavi (2022) – Assessing machine learning techniques in forecasting lumpy skin disease occurrence based on meteorological and geospatial features.

This study investigates the use of various machine learning algorithms to forecast the occurrence of LSD outbreaks in cattle using climatic, geospatial, and demographic data. The researchers first applied the Extra Trees Classifier for feature selection, identifying key predictors such as temperature, rainfall, animal population density, land cover, and elevation. They then trained multiple models—including Random Forests, SVM, and Artificial Neural Networks (ANNs)—to predict disease presence in unseen data. The results showed that ANNs performed best, achieving $AUC \approx 0.97$ and $F1\text{-score} \approx 0.94$, with overall accuracy around 97%. Interestingly, models using only meteorological features performed nearly as well as those including all data types, highlighting the dominant role of climate variables. The paper concludes that machine learning models, particularly ANNs, can support proactive surveillance and vaccination strategies in high-risk areas.

2.7 D.F. Dofadar, H.M. Abdullah, R.H. Khan, R. Rahman, M.S. Ahmed – IEEE ICAIET (2022) – A Comparative Analysis of Lumpy Skin Disease Prediction Through Machine Learning Approaches.

This study compares a range of traditional machine learning models such as Decision Trees, SVMs, Random Forests, and KNN for LSD outbreak prediction. They applied comprehensive feature selection and evaluated models on regional outbreak datasets, concluding that Random Forest and SVM provided the best balance of accuracy and interpretability. Hyperparameter tuning was employed to further boost model performance. Results showed high classification accuracy (> 95%) across models, with Random Forest excelling in handling class imbalance. The paper emphasizes its practical utility for real-time epidemiological monitoring, aiding decision-makers in veterinary health systems.

2.8 S. Tesfaye, F. Regassa, G. Beyene, S. Leta, J. Paeshuyse (Frontiers in Veterinary Science, 2024)- Spatiotemporal Analysis and Forecasting of LSD Outbreaks in Ethiopia.

This research utilizes time-series and spatial models, such as ARIMA and Bayesian methods, to analyze and predict LSD outbreaks in Ethiopian districts based on data from 2000 to 2015. The models successfully detected seasonal outbreak patterns and spatial hotspots, offering a data-driven forecast of high-risk periods and locations. Model validation was performed with hold-out and cross-validation techniques. The paper highlights the practical benefit of such forecasting tools for veterinary authorities, enabling more effective allocation of vaccines and resources. By predicting seasonal trends and outbreak hotspots, this method supports targeted prevention strategies to curb LSD spread.

CHAPTER 3

SOFTWARE REQUIREMENTS AND SPECIFICATIONS

SOFTWARE REQUIREMENTS AND SPECIFICATIONS

3.1 Functional Requirements

Functional requirements specify the core operations performed by the system. In this project, the system focuses on image-based disease prediction using Deep Machine Learning. It enables automated lesion detection and classification, allowing early identification of Lumpy Skin Disease. The WhatsApp interface ensures easy interaction for farmers and supports real-time diagnosis.

- The system shall accept cow skin images as input for disease analysis.
- The system shall preprocess images using image processing techniques.
- The system shall identify and segment lumpy skin lesions from the images.
- The system shall classify the images as Lumpy Skin Disease-affected or healthy using a trained DML model.
- The system shall display prediction results to users through a WhatsApp-based interface.

3.2 Non-Functional Requirements

Non-functional requirements define the quality and performance of the system. Accuracy and speed are critical for early disease detection. User-friendliness improves adoption among farmers, while scalability allows expansion to large-scale livestock monitoring. Security ensures safe communication and data protection in online platforms.

- The system should provide high accuracy in disease prediction.
- The system should deliver results with minimum response time.
- The system should be user-friendly for farmers with minimal technical knowledge.
- The system should be scalable to support large datasets and multiple users.

3.3 Software Requirements

The software requirements describe the technologies used to develop and deploy the system. Python is selected for its extensive machine learning support. PyTorch enables efficient model training and prediction, while Flask supports backend deployment. Twilio integrates WhatsApp-based communication, making the system practical and accessible for real-world farming environments.

- Programming Language: Python
- Deep Learning Frameworks: PyTorch, Torchvision, TIMM.
- Image Processing: Pillow.
- Scientific Libraries: NumPy, Scikit-learn.
- Visualization Tools: Matplotlib, Seaborn.
- Web Framework: FastAPI.
- Communication API: Twilio (WhatsApp Integration).

3.4 Hardware Requirements

The hardware requirements ensure smooth execution of deep learning and image processing tasks involved in the project. Adequate RAM and processing power are essential for training and testing the DML model efficiently. GPU support significantly accelerates model training, especially for large image datasets. Sufficient storage is required for handling datasets and trained models, while an internet connection enables cloud communication and WhatsApp-based disease prediction

- A computer or laptop with a minimum of 8 GB RAM.
- Intel i5 / AMD Ryzen 5 processor or higher.

- Minimum 20 GB free storage space for datasets and model files.
- Stable internet connection for dataset access and WhatsApp API communication.
- Smartphone with WhatsApp for farmer-side image submission.

CHAPTER 4

SYSTEM ANALYSIS AND DESIGN

SYSTEM ANALYSIS AND DESIGN

The goal of system analysis and design is to comprehend how internal processes are structured and how the Lumpy Skin Disease (LSD) prediction system functions from the viewpoint of the user.

In order to produce precise disease predictions, this section describes who utilizes the system, what is done, and how data is processed step-by-step.

Disease detection may be carried out with little technical expertise thanks to the design's simplicity, efficiency, and friendliness to farmers.

4.1 Use Case Diagram

The Use Case Diagram represents the functional behavior of the LSD Prediction System and clearly shows the interaction between the farmer and the system.

In this system:

1. The Farmer is the primary user who interacts with the application.
2. The farmer uploads images of cows suspected of having Lumpy Skin Disease.
3. The system processes the uploaded image and performs disease prediction using a deep learning model.
4. After prediction, the farmer can view the results and receive suitable recommendations.

The diagram highlights the core services provided by the system, such as image upload, disease prediction, result visualization, and recommendation generation.

This diagram helps in understanding what the system does, without showing internal technical details.

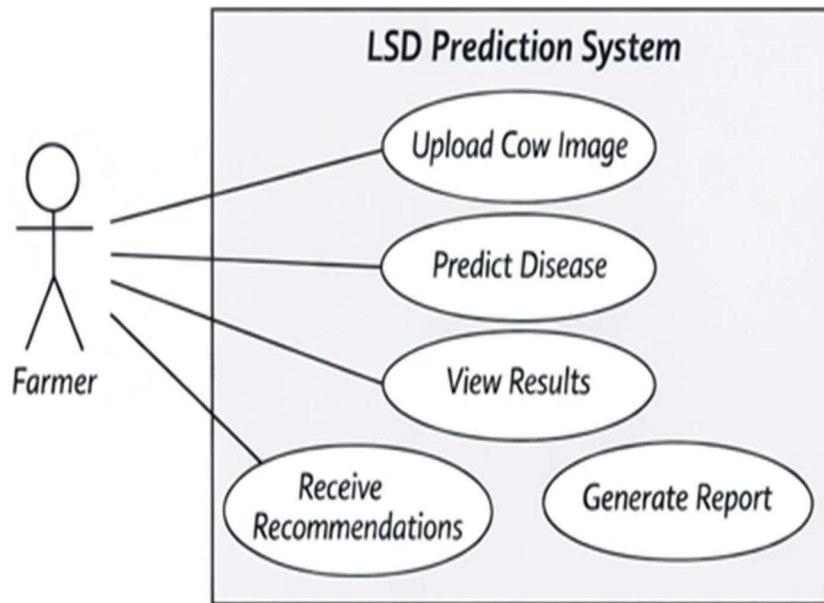


Fig 4.1: Use Case Diagram for LSD Prediction System.

4.2 Activity Diagram

The Activity Diagram explains the complete workflow of the disease prediction process in a sequential manner.

It describes how the system operates internally once the farmer interacts with it. The process begins when the farmer uploads a cow image.

The system then preprocesses the image to improve prediction accuracy.

Next, the deep machine learning model analyzes the image to identify whether the cow is affected by Lumpy Skin Disease.

If the disease is detected, the system evaluates the severity level.

Based on the prediction and severity, appropriate recommendations are generated and displayed to the farmer.

This diagram clearly shows the flow of actions from start to end, making it easy to understand how input is converted into useful output.

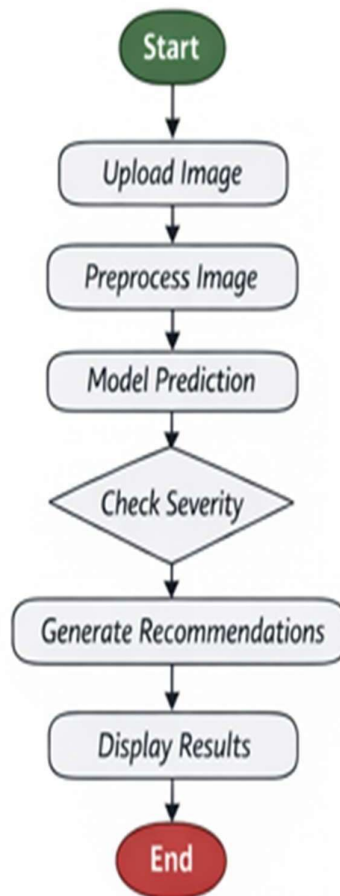


Fig 4.2: Activity Diagram of Disease Prediction Process.

CHAPTER 5

IMPLEMENTATION

IMPLEMENTATION

5.1 Introduction

One of the most important phases in the creation of the Lumpy Skin Disease (LSD) Prediction System is the implementation phase, which transforms the theoretical concepts and suggested system architecture into a fully functional, real-time application. The implementation phase concentrates on the actual development, integration, and execution of the system components, whereas earlier stages like system analysis and design concentrate on comprehending requirements and creating workflows.

The implementation of this project entails creating a Deep Machine Learning-based disease detection system that can use image data to identify Lumpy Skin Disease in cows. Early illness detection, which is essential for lowering cattle mortality, minimizing financial losses, and enhancing animal health management, is supported by the system's design. To ensure consistency between design and execution, the implementation closely adheres to the workflows shown in the Use Case Diagram and Activity Diagram.

Each module of the LSD Prediction System is in charge of a distinct function, such as image acquisition, preprocessing, disease prediction, severity analysis, recommendation generation, and user interaction. The system is designed in a modular and tiered manner. Even when the project complexity rises, this methodical approach guarantees that the system will continue to be scalable, maintainable, and simple to comprehend. Smooth data flow from input to output is made possible by each module communicating with other modules in a predetermined order.

The incorporation of deep learning algorithms for precise image-based disease prediction is a primary priority during deployment. Large-scale livestock monitoring is not feasible with traditional manual inspection techniques since they take a lot of time and require specialized skills. A Vision Transformer (ViT)-based deep learning model, which automatically learns intricate visual patterns such skin lesions, nodules, swelling, and textural differences linked to Lumpy Skin Disease, is used by the implemented system to get over these restrictions.

Even with little training data, the system can achieve high accuracy by using a pretrained model in conjunction with fine-tuning.

Effective use of resources is another crucial component of implementation. Because the system is meant to be implemented in real-world settings, particularly in rural and low-resource settings, extra attention is taken to guarantee that the model can function on systems with constrained hardware capabilities. To lower memory use and computational cost, strategies such as gradient accumulation, mixed precision training, and selective layer fine-tuning are used. Because of this, the technology is feasible and reasonably priced for broad use.

Another crucial element of implementation is data management. The system is made to keep timestamps, severity levels, prediction results, and uploaded photos in an organized way. This makes it possible to track past illnesses, aids in tracking cattle health over time, and promotes future model advancement. To guarantee system dependability and stop inaccurate predictions brought on by invalid inputs, appropriate error management and validation procedures are also put in place.

5.2 System Implementation Architecture

The Lumpy Skin Disease (LSD) Prediction System's structure and component interactions for effective disease identification are explained in the System Implementation Architecture. Every step of the prediction process is executed smoothly thanks to the architecture's modular, scalable, and easily maintained design.

The input and interaction layer, the processing and prediction layer, and the output and storage layer are the three primary layers of the system's layered architecture. This layer separation facilitates system functionality organization and enhances clarity during deployment and future improvement.

The system's entrance point is the Input and Interaction Layer. Farmers use a WhatsApp-based interface to communicate with the system in this experiment, uploading pictures of cows they believe may have lumpy skin disease.

Because farmers may utilize a well-known communication platform without installing specialist apps, this method makes the system very accessible. Before sending the image to the processing layer, the system verifies that the format and quality are right.

The core component of the system architecture is the Processing and Prediction Layer. In order to prepare the image for deep learning analysis, it is first subjected to image preprocessing, which includes scaling, normalization, and transformation. Following preprocessing, the image is sent into a deep machine learning model based on Vision Transformer (ViT), which extracts significant visual elements such as nodules, skin lesions, and textural differences. The model predicts whether the cow is healthy or impacted by LSD based on these learning features.

The approach uses severity analysis to categorize the infection as mild, moderate, or severe if the disease is found. Instead of only producing a binary output, this stage deepens the prediction process and facilitates better treatment choices. Data storage and results delivery to the farmer are handled by the Output and Storage Layer. The farmer receives the prediction result, severity level, and suggested actions via WhatsApp in an easy-to-understand manner. The system also keeps forecast information in a database, including disease state, severity, and timestamp. Future analysis and historical tracking are supported by this saved data.

The LSD Prediction System is appropriate for practical agricultural usage since the system implementation architecture guarantees effective data flow, distinct responsibility separation, and dependable operation.

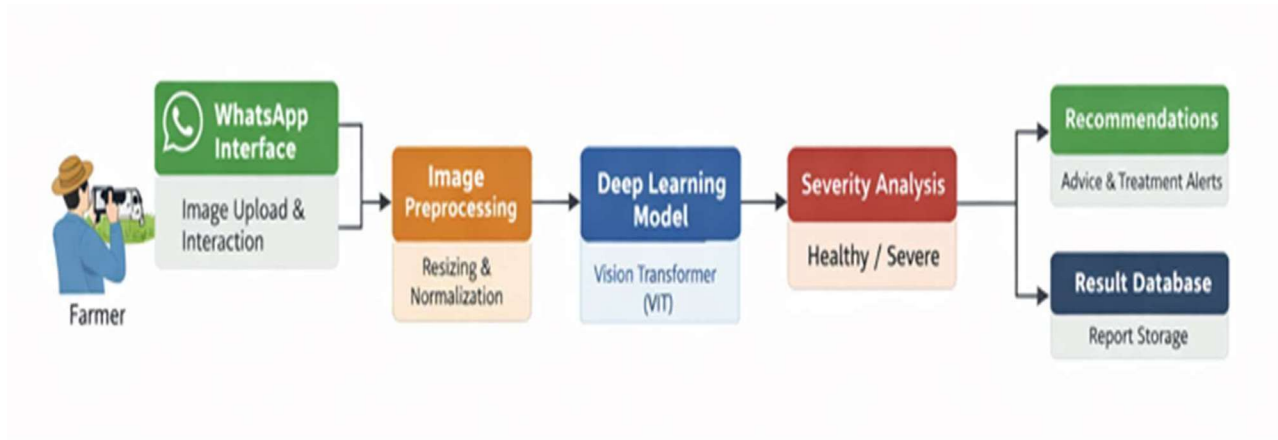


Fig 5.2: System Implementation Architecture.

5.3 Algorithm for Disease Prediction

The proposed system uses a Vision Transformer (ViT)–based Deep Machine Learning algorithm to predict Lumpy Skin Disease in cows using image data. The algorithm follows a structured pipeline consisting of image acquisition, preprocessing, deep feature extraction, classification, severity estimation, and recommendation generation.

The algorithm is designed to work efficiently on low-resource systems while maintaining high prediction accuracy.

Algorithm: LSD Prediction Using Vision Transformer.

Input: Cow skin image uploaded by the farmer.

Output: Disease status (Normal / LSD Affected), severity level, and recommendations.

Step 1: Image Input

The farmer uploads an image of the cow through the user interface or WhatsApp-based system.

Step 2: Image Validation

The system checks whether the uploaded file is a valid image format and meets minimum quality requirements.

Step 3: Image Preprocessing

Prediction of lumpy skin disease in cows using dml

Resize image to a fixed resolution

Normalize pixel values

Apply necessary transformations

Step 4: Feature Extraction Using Vision Transformer

The preprocessed image is divided into patches Each

patch is embedded into a vector representation

Transformer encoder layers learn global and local image features

Step 5: Disease Classification

The model outputs a probability score using a sigmoid activation function

If probability < threshold $\rightarrow$ Healthy

If probability $\geq$ threshold $\rightarrow$ LSD detected

Step 6: Severity Estimation

Severity score is computed based on prediction confidence

Severity is classified as Mild, Moderate, or Sever

Step 7: Recommendation Generation

Based on severity, suitable treatment or advisory messages are generated

Step 8: Result Display and Storage

Results are displayed to the farmer

Prediction details are stored in the database

End.

5.4 Dataset Collection and Organization

The dataset plays a critical role in determining the accuracy and reliability of the deep learning model used in the Lumpy Skin Disease (LSD) Prediction System. Since the system is based on image analysis, the quality, diversity, and organization of the dataset directly influence the model's ability to distinguish between healthy cows and LSD-affected cows. In this project, the dataset consists of digital images representing both normal cow skin and cow skin infected with Lumpy Skin Disease. These images capture visible symptoms such as nodules, lesions, swelling, and texture variations, which are essential indicators for disease prediction.

The dataset is designed to include images taken under different environmental conditions, lighting variations, and viewing angles. This diversity helps the model learn robust visual patterns and improves its generalization capability when predicting disease from unseen images. Proper dataset organization ensures smooth integration with preprocessing, training, and prediction modules, thereby improving overall system performance.

The foundation of any successful machine learning project is the quality and organization of the dataset, particularly when dealing with image-based disease detection. In this project, the dataset consists of images of cow skin, categorized into healthy cows and cows infected with Lumpy Skin Disease (LSD). The process of collecting, verifying, and organizing these images is critical to ensure the model learns meaningful patterns and achieves accurate predictions.

5.4.1 Data Sources

A diverse and representative dataset is essential for the model to generalize well. For this project, images of cow skin were collected from multiple sources.

Field Collection from Farms and Veterinary Clinics: Real-world images were captured directly from cows on farms and in clinics with varying environmental conditions. This approach ensured inclusion of natural variations in lesion size, texture, shape, and location, which are typical in LSD.

The images were captured using high-resolution cameras under different lighting conditions to mimic real-world scenarios.

Publicly Available Datasets: To supplement real-world images, open-source datasets containing livestock skin images were included. This helped cover additional breeds, geographical variations, and different stages of the disease.

Synthetic Data via Augmentation: To enhance dataset diversity and address class imbalance, data augmentation techniques were applied. These included:

Rotation: Randomly rotating images to simulate different orientations.

Flipping: Horizontal and vertical flips to ensure the model learns orientation-invariant features.

Scaling and Cropping: Adjusting image size to simulate distance variations during photography.

Brightness and Contrast Adjustment: Modifying lighting to improve robustness against different environmental conditions.

Noise Injection: Adding slight noise to simulate real-world imperfections in images.

This multi-source approach ensured the dataset captured a wide range of real-life scenarios, improving the generalization ability of the model.

5.4.2 Data Labeling and Annotation

Labeling images accurately is fundamental for supervised learning. Each image was annotated into one of two classes.

Normal (Healthy): No visible lesions, nodules, or symptoms of LSD.

Infected (Lumpy Skin Disease Positive): Showing one or more lesions, nodules, or characteristic skin bumps.

Veterinary professionals were consulted to verify all annotations. This expert validation minimized the risk of mislabeled images, which could mislead the model and reduce predictive accuracy. Additionally, a structured naming convention was used (e.g., Normal_001.jpg, Infected_001.jpg) to maintain traceability and organization.

5.4.3 Dataset Organization and Structure

Proper organization of the dataset is crucial for efficient model training and evaluation. The following steps were implemented:

Folder-Based Organization: The dataset was organized into main folders corresponding to **labels:** Normal and Infected. Within each folder, images were stored sequentially for easy access.

Standardization of Image Size: All images were resized to 224x224 pixels, which is compatible with the Vision Transformer (ViT) model used in this project. This ensured uniform input dimensions, reducing computational complexity and preventing shape-related errors during model training.

Normalization of Pixel Values: Image pixel values were normalized to a 0–1 range, improving model convergence and training stability.

Dataset Splitting: The dataset was split into training (80%) and validation (20%) sets. The training set enables the model to learn disease patterns, while the validation set assesses model performance on unseen data, helping detect overfitting and underfitting.

Class Balancing: Since infected images were fewer than healthy ones, oversampling via augmentation ensured the model did not become biased towards the majority class, which is critical for accurate disease detection.

5.4.4 Preprocessing Challenges:

Preparing a high-quality dataset posed several challenges:

Variability in Lesion Appearance: Lesions differ in size, shape, and color, making consistent labeling difficult. Close collaboration with veterinarians was necessary to ensure correct annotation.

Lighting and Background Variations: Images captured in farms had inconsistent lighting and cluttered backgrounds, which could introduce noise. Preprocessing techniques such as histogram equalization and background filtering were applied to minimize this effect.

Limited Infected Samples: Acquiring images of infected cows was challenging due to the lower prevalence of LSD. Data augmentation helped address this limitation, providing enough diverse samples for training.

Image Quality Variations: Some images were blurred or low-resolution. Such images were either enhanced or excluded to maintain dataset quality.

5.4.4 Significance of an Organized Dataset

An organized and high-quality dataset is the backbone of machine learning performance. Proper collection, labeling, augmentation, and preprocessing allow the model to:

Learn relevant features of LSD lesions accurately.

Generalize well to new, unseen images.

Reduce misclassification and prediction errors.

Improve training efficiency and reduce computational time.

By establishing a structured workflow for dataset preparation, this project ensures that the subsequent steps of model training, validation, and testing are built on a solid foundation, resulting in a reliable predictive system for early detection of Lumpy Skin Disease in cows.

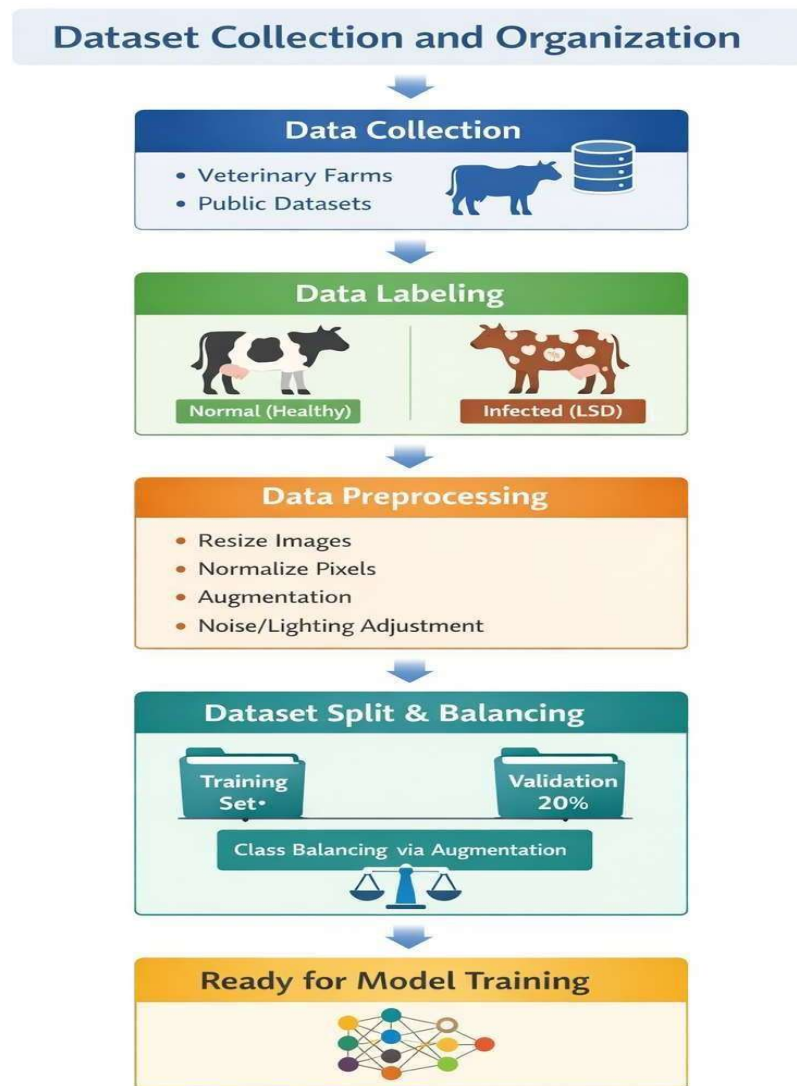


Fig 5.4: Dataset Collection and Organization Flowchart

5.5 Image Acquisition Module

The Image Acquisition Module is a critical component of the Lumpy Skin Disease (LSD) prediction system. It serves as the interface through which farmers provide input images of cows for disease detection. This module ensures that the system receives high-quality, validated images suitable for processing by the deep learning model.

5.5.1 Purpose of the Module

The primary goal of the Image Acquisition Module is to enable farmers or veterinary personnel to submit images of cows' skin safely and efficiently. Since the predictive accuracy of the system heavily depends on the quality of input images, this module plays a crucial role in capturing usable and meaningful data.

Key objectives include:

- Ensuring ease of use for non-technical users like farmers.
- Supporting real-time image upload from mobile devices or cameras.
- Validating images to ensure they meet the system's minimum quality requirements.

5.5.2 Key Functions

The module performs several important functions:

1. Image Input from Multiple Sources:
 - Accepts images captured via smartphone cameras, digital cameras, or other mobile devices.
 - Provides a user-friendly interface for selecting and uploading images.
2. Support for Common Image Formats:
 - Compatible with standard formats such as JPEG, PNG, and BMP.
 - Ensures flexibility and convenience for farmers using different devices.

3. Image Validation:

- Checks the size, resolution, and format of uploaded images.
- Ensures that the images are clear, well-lit, and focused on the cow's skin, reducing errors in subsequent preprocessing and prediction.
- Rejects unsupported formats or low-quality images and provides feedback to the user.

5.6 Image Preprocessing Module

The Image Preprocessing Module is a crucial step in the Lumpy Skin Disease (LSD) prediction system. Its primary purpose is to prepare raw images for deep learning model input by improving their consistency, quality, and usability. Raw images captured from different sources, such as farms or veterinary clinics, often vary in size, resolution, lighting conditions, and orientation. Preprocessing ensures that these images are standardized and optimized for model training and prediction, which directly affects the accuracy and reliability of the disease detection system. This module corresponds to the Preprocess Image activity in the system's Activity Diagram, representing the transformation of uploaded raw images into model-ready inputs.

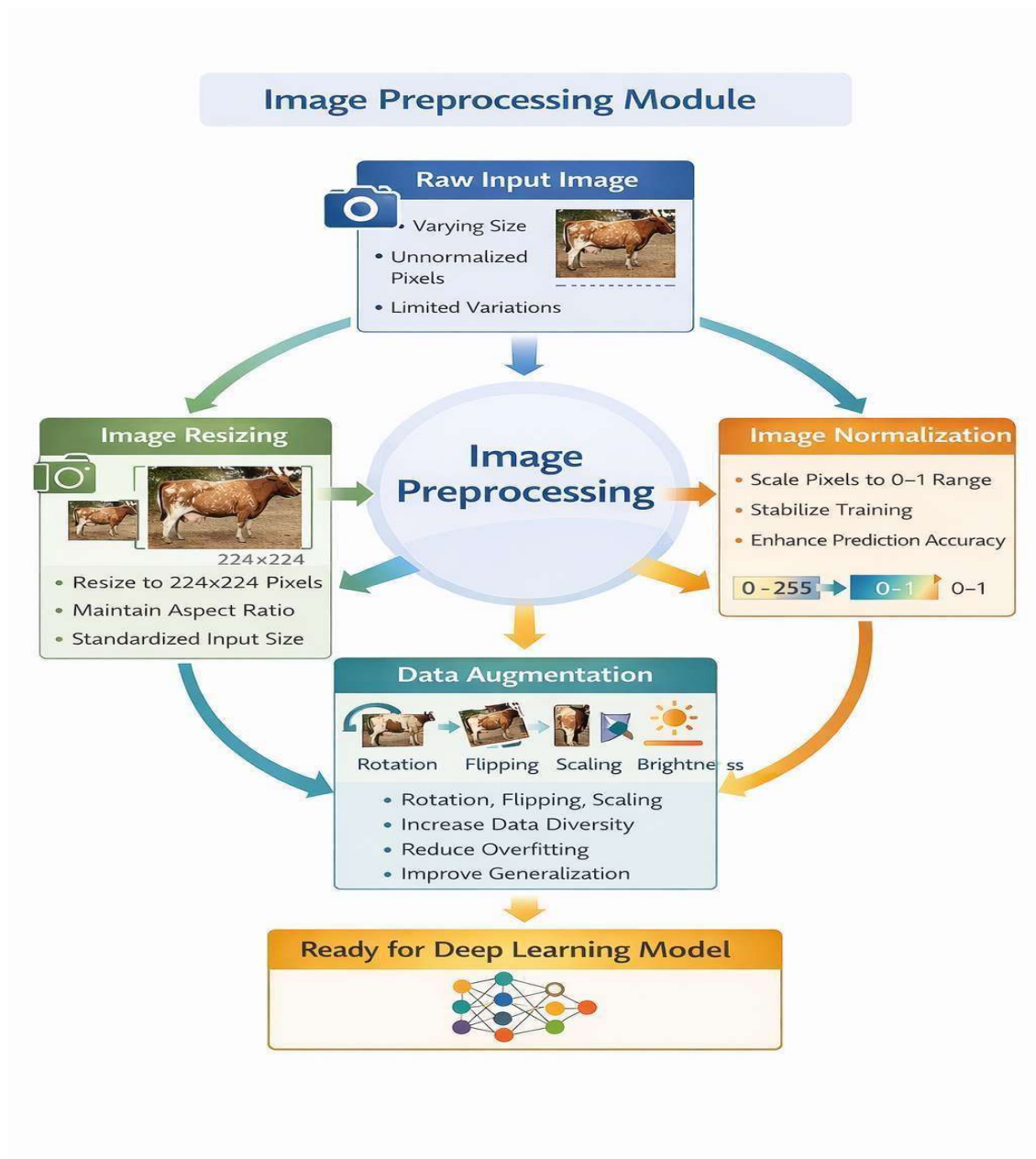


Fig 5.6: Image Preprocessing Module.

5.6.1 Image Resizing

Raw images from various devices have different dimensions and aspect ratios, which can cause inconsistencies during model training. To standardize the input:

- **Fixed Dimensions:** All images are resized to a specific size, typically 224x224 pixels, which is compatible with the Vision Transformer (ViT) model.
- **Aspect Ratio Maintenance:** To avoid distortion, resizing is often combined with padding or cropping, preserving the visual features of the cow's skin and lesions.
- **Uniform Processing:** Standardized image dimensions allow the model to process all images consistently, ensuring accurate feature extraction.

Benefits of Image Resizing:

- Reduces computational requirements and memory usage.
- Ensures consistent input for training and prediction.
- Improves the efficiency of feature learning by focusing on relevant areas of the image.

5.6.2 Image Normalization

Normalization is applied to ensure that the pixel intensity values are consistent across all images:

- **Scaling Pixel Values:** Original pixel values range from 0–255. Normalization scales these values to a 0–1 range, which is more suitable for deep learning models.
- **Improved Training Stability:** Normalized inputs reduce the risk of gradient explosion or vanishing during training, leading to faster convergence.
- **Enhanced Prediction Accuracy:** Standardized values make the model less sensitive to variations in lighting and intensity, improving generalization on unseen data.

Benefits of Image Normalization:

- Accelerates model training.
- Reduces bias from differences in image capture conditions.
- Ensures consistency across datasets from multiple sources.

5.6.3 Data Augmentation Techniques

Data augmentation increases the diversity of the dataset and helps the model generalize better:

➤ Techniques Used: Rotation, flipping (horizontal and vertical), scaling, and minor cropping.

Purpose:

- **Increase Dataset Diversity:** Provides varied examples from existing images, reducing the need for collecting excessive new data.
- **Reduce Overfitting:** Prevents the model from memorizing training images by exposing it to slightly altered variations.
- **Improve Generalization:** Enhances the model's ability to correctly predict LSD in new, unseen images, even if captured under different conditions.

By applying augmentation, the system ensures that the deep learning model learns robust features that are representative of real-world variations in cow skin appearance.

5.7 Deep Machine Learning Model Implementation

The Deep Machine Learning (DML) Model Implementation is the core of the LSD prediction system. It involves selecting an appropriate model, designing its architecture, and optimizing it to detect Lumpy Skin Disease in cow images with high accuracy. This section describes the choices and strategies used for model selection, architecture design, and transfer learning to achieve reliable predictions.

5.7.1 Model Selection

Selecting the right deep learning model is a critical step in ensuring high accuracy, efficiency, and robustness in disease prediction. For this project, two model families were considered: Vision Transformers (ViT) and Convolutional Neural Networks (CNNs). Both have proven effectiveness in image classification tasks, but their advantages differ based on dataset size, complexity, and feature representation.

- **Vision Transformer (ViT):**

ViTs divide an image into small patches, encode these patches into vectors, and process them through self-attention layers. This architecture allows the model to capture long-range dependencies and global context, which is important for detecting subtle patterns in skin lesions caused by LSD. Unlike traditional CNNs, which rely on local convolution filters, ViTs analyze the entire image holistically, improving sensitivity to complex lesion patterns that may vary in size, shape, and distribution.

- **Convolutional Neural Network (CNN):**

CNNs apply convolutional filters to extract spatial features from images, followed by pooling layers that reduce dimensionality while retaining important features. CNNs have a proven track record in medical and agricultural image classification tasks. They are especially effective when combined with transfer learning, using pre-trained models such as ResNet, VGG, or Efficient Net to leverage learned feature representations from large image datasets.

Rationale for Selection:

For this project, ViT was selected due to its ability to model global interactions between patches, which is crucial for LSD detection because lesions can appear in different locations and sizes across the cow's body. The ViT model is also compatible with transfer learning, allowing us to leverage pre-trained weights and reduce training time without compromising accuracy.

Key Advantages of the Selected Model:

1. High Accuracy: Can capture subtle and complex patterns in images.
2. Flexibility: Handles varying lesion locations and sizes.
3. Scalability: Can be fine-tuned for larger datasets in the future.
4. Reduced Overfitting Risk: Self-attention layers in ViT reduce the model's tendency to overfit on small datasets compared to deep CNNs with many parameters.

5.7.2 Model Architecture Design

The architecture of the deep learning model was carefully designed for binary classification (Normal vs LSD) while maintaining efficiency and robustness. The ViT-based model is composed of several key layers:

1. Input Layer:

- Accepts preprocessed images of size 224x224 pixels with three color channels (RGB).
- The input is split into 16x16 patches, which are flattened and transformed into patch embeddings for the transformer.

2. Patch Embedding Layer:

- Converts image patches into vector embeddings using a linear projection.
- Adds positional encoding to retain spatial information across patches.
- This allows the model to recognize the position of lesions, which is critical for accurate LSD detection.

3. Transformer Encoder Layers:

- Composed of multi-head self-attention blocks and feed-forward networks.
- Self-attention layers capture global context, allowing the model to analyze relationships between patches.
- Feed-forward layers refine feature representations for better classification.

4. Classification Head:

- A fully connected layer outputs a probability score for each class (Normal vs LSD).
- Uses a softmax activation function to determine the final prediction.

5. Optimization for Binary Classification:

- Loss function: Binary Cross-Entropy, optimized for two-class prediction.
- Additional dropout layers are included to prevent overfitting and improve generalization.

Benefits of This Architecture:

- * Efficiently handles high-resolution images.
- * Extracts both local and global features of skin lesions.
- * Reduces model complexity while maintaining accuracy.
- * Allows easy integration of transfer learning for improved performance.

5.7.3 Transfer Learning and Fine-Tuning Strategy

Transfer learning is employed to leverage pre-trained weights from large-scale image datasets, such as ImageNet, to improve model performance on the LSD dataset. This approach significantly reduces training time and enhances accuracy, especially when the available dataset is limited.

Strategy Used:

1. Loading Pre-trained Weights:

- The ViT model is initialized with weights from a model trained on a large general dataset.
- Early layers retain feature detectors for basic visual patterns such as edges, textures, and shapes.

2. Layer Freezing:

- Initial layers are frozen, preventing their weights from being updated during training.
- This allows the model to retain general visual knowledge while focusing on learning LSD- specific patterns.

3. Fine-Tuning Selected Layers:

- Later layers and the classification head are fine-tuned on the LSD dataset.
- Fine-tuning allows the model to adapt to domain-specific features such as lesion size, distribution, and color variations.

4. Benefits of Transfer Learning:

- Reduces training time and computational resources.
- Improves convergence and model stability.
- Enhances accuracy, especially when the dataset is small or moderately sized.
- Mitigates the risk of overfitting while leveraging powerful pre-learned representations.

Conclusion:

The combination of ViT model selection, carefully designed architecture, and transfer learning provides a robust foundation for accurate Lumpy Skin Disease prediction. The model can detect subtle lesion patterns, generalize across different images, and efficiently classify cows as healthy or infected, forming the core of the intelligent disease prediction system.

5.8 Model Training Process

The Model Training Process is the stage where the deep learning model learns to predict Lumpy Skin Disease (LSD) from cow images. During training, the model adjusts its internal parameters (weights and biases) based on the input data and the difference between predicted and actual labels. Proper training is crucial to achieve high accuracy and robust predictions while avoiding overfitting or underfitting.

5.8.1 Training–Validation Split

Before training begins, the dataset is divided into training and validation sets:

➤ **Training Set (typically 80%):**

- Used to train the model by updating its weights iteratively.
- The model learns patterns, textures, and features associated with healthy and infected cow skin.

➤ **Validation Set (typically 20%):**

- Used to evaluate the model's performance on unseen data during training.
- Helps in monitoring overfitting; if training accuracy is high but validation accuracy stagnates or drops, it indicates the model is memorizing rather than generalizing.

Importance of the Split:

- Ensures that the model is tested on independent data that it has not seen during training.
- Provides metrics to adjust hyperparameters such as learning rate, batch size, and number of epochs.
- Forms the foundation for later testing and deployment stages by showing how well the model generalizes to new images.

5.8.2 Loss Function and Optimization

Loss functions and optimizers are critical components that guide the model to learn correctly:

➤ **Loss Function:**

- Measures the difference between the predicted output and the actual label.
- In this project, Binary Cross-Entropy Loss is used since the problem is binary classification (Normal vs LSD).

- Minimizing this loss ensures the model predicts the correct class with high confidence.

➤ **Optimization Algorithm:**

- Optimizers adjust model weights to minimize the loss function.
- Adam Optimizer is commonly used due to its adaptive learning rate, fast convergence, and stability.
- Gradient descent updates the model parameters in the direction that reduces error, iteratively improving prediction accuracy.

Importance of Loss and Optimization:

- Determines how efficiently the model learns from the dataset.
- Influences convergence speed and final accuracy.
- Ensures stable learning without oscillations or divergence, particularly in deep models like Vision Transformers.

5.8.3 Hardware and Resource Optimization

Training deep learning models can be computationally expensive, especially on systems with limited resources like low VRAM GPUs or standard CPUs. To address this, several optimization techniques are applied:

1. Gradient Accumulation

- Allows training with a larger effective batch size than the hardware can handle in memory.
- Gradients are accumulated over several mini-batches before updating weights, reducing memory requirements while preserving training stability.

2. Mixed Precision Training:

- Combines 16-bit and 32-bit floating point calculations to speed up training and reduce memory usage.
- Particularly effective on GPUs supporting tensor cores, as it accelerates computation without significant loss of accuracy.

5.9 Disease Prediction Module

The Disease Prediction Module is the most critical operational component of the proposed Lumpy Skin Disease (LSD) detection system. This module is responsible for using the trained deep machine learning model to analyze cow skin images and determine whether the animal is healthy or affected by Lumpy Skin Disease. It transforms learned knowledge from the training phase into practical, real-world disease diagnosis.

Once the deep learning model has completed training and validation, it is deployed in inference mode within this module. The primary objective of this module is to provide accurate, fast, and interpretable predictions that can support farmers and veterinarians in early disease identification and timely intervention.

Input Processing for Prediction

The Disease Prediction Module receives input images from the Image Acquisition Module, typically uploaded by farmers using mobile phones or cameras. These images may vary in lighting, resolution, and orientation. Before prediction, the images are passed through the Image Preprocessing Module, where the following operations are performed:

- Image resizing to a fixed dimension compatible with the model
- Pixel value normalization for consistent intensity levels
- Optional augmentation during testing for robustness.

Once preprocessing is completed, the image is converted into a tensor format and forwarded to the trained deep learning model.

Model Inference Mechanism

During inference, the trained model processes the image through multiple internal layers:

➤ Feature Extraction:

The model extracts key visual patterns such as skin texture, presence of nodules, lesion boundaries, swelling, and abnormal color distribution.

➤ Pattern Analysis:

Deep layers analyze spatial and contextual relationships across the image to distinguish between normal skin and disease-affected regions.

Classification Decision:

The final classification layer outputs probabilities corresponding to two classes:

Prediction of lumpy skin disease in cows using dml

1. Healthy
2. LSD-Affected

The class with the highest probability is selected as the final prediction.

Disease Classification Output

The Disease Prediction Module categorizes the input image into one of the following classes:

1. Healthy
 - Indicates the absence of visible LSD symptoms.
 - Suggests normal skin texture and color patterns.
 - No immediate veterinary action is required, though periodic monitoring is recommended.
2. LSD-Affected
 - Indicates the presence of visual characteristics associated with Lumpy Skin Disease.
 - Suggests the need for veterinary confirmation, isolation of the animal, and treatment.

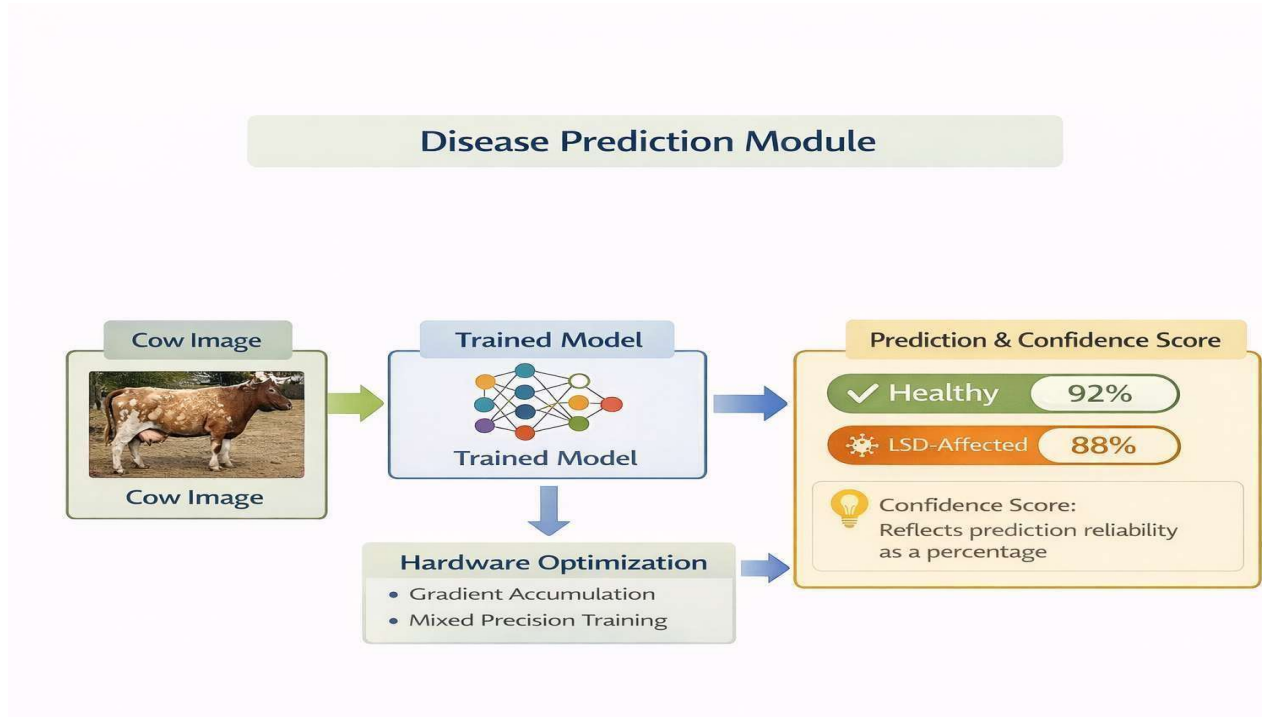


Fig 5.9: Disease Prediction Module

5.10 Severity Analysis Module

The Severity Analysis Module is an advanced decision-support component of the Lumpy Skin Disease (LSD) prediction system. While the Disease Prediction Module identifies whether a cow is healthy or affected by LSD, the Severity Analysis Module further evaluates how serious the disease condition is. This additional analysis helps farmers and veterinarians understand the urgency of medical intervention and prioritize treatment effectively.

This module is activated only when LSD is detected by the Disease Prediction Module. If the cow is classified as healthy, the severity analysis is skipped, ensuring efficient system execution. Input to the Severity Analysis Module. The module receives the following inputs from the Disease Prediction Module:

- Predicted disease label (Healthy or LSD-Affected).
- Prediction probability / confidence score generated by the deep learning model

The confidence score represents the model's certainty that the cow is affected by LSD. This probability value forms the basis for severity evaluation

Severity Evaluation Mechanism

The Severity Analysis Module interprets the prediction probability to assess the level of disease severity. Instead of performing a separate medical diagnosis, this module provides an AI- assisted estimation of severity based on visual symptom intensity detected in the image.

The logic works as follows:

- If the prediction label is Healthy, the severity level is automatically marked as Healthy.
- If the prediction label is LSD-Affected, the confidence score is analyzed to determine severity.

Higher confidence scores typically indicate:

- Clear and prominent skin lesions.
- Multiple nodules.
- Strong visual patterns associated with LSD.

Lower confidence scores may indicate:

- Early-stage symptoms.
- Mild or unclear lesions.
- Poor image quality.

Severity Categories

The system classifies severity into the following categories:

1. Healthy

- No signs of LSD detected.
- Prediction confidence indicates normal skin condition.
- No immediate action required.
- Routine monitoring recommended.

2. Severe

- High prediction confidence indicating strong presence of LSD symptoms.
- Visible nodules, swelling, or abnormal skin texture.
- Immediate veterinary attention recommended.
- Isolation and treatment may be necessary to prevent spread.

5.11 User Interaction Module

The User Interaction Module is a vital component of the Lumpy Skin Disease (LSD) prediction system, designed to ensure smooth and effective communication between the intelligent backend system and the end users, primarily farmers and veterinary professionals. Since the target users may not have advanced technical knowledge, this module emphasizes simplicity, accessibility, and clarity. Its core function is to present complex deep learning results in an easy-to-understand and actionable format.

This module ensures that the benefits of artificial intelligence and deep machine learning are delivered directly to rural communities in a practical and usable manner.

5.11.1 WhatsApp-Based Interaction System

The system uses a WhatsApp-based chatbot to enable seamless interaction. WhatsApp was selected because it is widely used, reliable, and familiar to farmers, especially in rural areas.

Key Functional Features

1. Image Submission via Chat

- Farmers can send cow images directly through WhatsApp chat.
- The bot automatically detects incoming images and initiates analysis.

2. Automated Message Handling

- The chatbot responds instantly with acknowledgment messages such as “Image received successfully.”
- This confirms successful submission and improves user confidence.

3. Guided Interaction Flow

- The bot provides simple instructions like:
- “Please upload a clear image of the cow’s skin.”
- “Ensure good lighting for better results.”

4. Multilingual Support (Optional Enhancement)

- The system can be extended to support regional languages, making it more inclusive.
- This significantly improves adoption among farmers who may not be fluent in English.

5. Low Bandwidth Compatibility

- WhatsApp works effectively even on low internet connections, ensuring uninterrupted service.

6. Secure Communication

- Messages and images are handled securely, ensuring user privacy and data protection.

5.11.2 Result Communication to Farmer

Once the image is processed, the system communicates the results directly to the farmer via WhatsApp messages.

Information Delivered to the Farmer.

1. Disease Status

- Clearly states whether the cow is Healthy or LSD-Affected.
- Avoids technical terms to ensure easy understanding.

2. Severity Level

- Indicates whether the condition is Healthy or Severe.
- Helps farmers assess urgency and prioritize treatment.

3. Actionable Recommendations

Provides clear and practical steps based on the result, such as:

- Isolate infected animals.
- Improve sanitation and hygiene.
- Consult a veterinarian immediately.
- Monitor symptoms regularly.

4. Confidence and Clarity

- Results are presented in a structured format using symbols or bullet points for readability.

5.12 Scalability

Scalability and future enhancement are essential considerations for any intelligent agricultural system intended for real-world deployment. The proposed Lumpy Skin Disease (LSD) Prediction System using Deep Machine Learning is designed with a scalable architecture that allows it to grow beyond a single user, device, or farm. This section discusses how the system supports expansion and outlines possible improvements that can enhance accuracy, usability, and impact in the future.

Scalability of the Proposed System

Scalability refers to the system's ability to handle increasing users, data volume, and computational load without degrading performance. The modular design of the system ensures easy extension and maintenance.

1. Mobile App Integration

Although the current system uses a WhatsApp-based interface for accessibility, it is designed to support dedicated mobile application integration in the future.

2. Real-Time Disease Detection

The system is optimized to support real-time or near real-time disease detection, ensuring immediate feedback after image upload.

3. Cloud Deployment

The system can be deployed on cloud platforms, enabling scalable storage, computation, and accessibility.

4. Multi-Farm Support

The system is designed to support multiple farms and farmers simultaneously, making it suitable for large-scale agricultural deployment.

Scalability Benefits:

- Supports regional, district-level, or national deployment.
- Enables centralized monitoring of disease outbreaks.
- Facilitates data-driven agricultural planning.

CHAPTER 6

RESULTS AND ANALYSIS

6.1 Experimental setup

The environment used to test and implement the suggested system is described in this section. Transparency and reproducibility of the results are guaranteed by a thorough description of the experimental setup.

The solution is appropriate for real-world implementation because the trials were carried out on a system with constrained hardware resources. A dataset of cow photos, including both healthy and LSD-affected animals, was used to train and evaluate the deep learning model.

The software environment included:

Programming language: Python

Deep learning framework: TensorFlow / PyTorch Supporting
libraries for image processing and data handling

The dataset was divided into training and validation sets, and unseen images were used during testing to evaluate real-time prediction capability.

6.2 Evaluation Metrics

To assess the performance of the proposed system, standard evaluation metrics were used. These metrics help measure how accurately the model classifies cow images.

The following metrics were considered:

Accuracy: Measures overall correctness of predictions Precision:

Measures correctness of LSD-positive predictions Recall:

Measures the model's ability to detect LSD cases

F1-score: Balances precision and recall.

Confusion Matrix: Shows correct and incorrect classifications

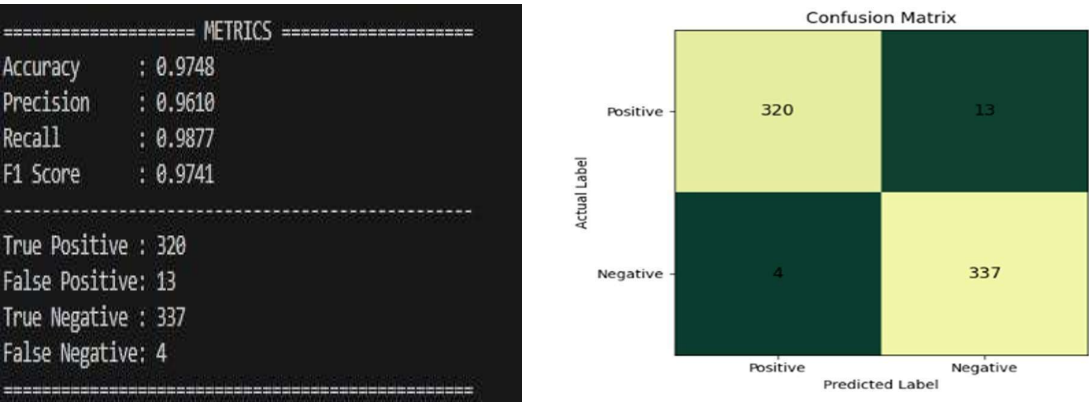


Fig 6.2: Evaluation Metrics.

6.3 Model Performance Results

This section presents the quantitative results achieved by the deep learning model during training and testing.

The model demonstrated strong performance in distinguishing between healthy and LSD- affected images. Training and validation accuracy showed consistent improvement over epochs, indicating effective learning. Loss values decreased steadily, confirming convergence of the model.

The optimized training strategy, including transfer learning and hardware optimization, contributed to improved accuracy even on low-resource systems.

6.4 Sample Prediction Results

This section showcases the system’s prediction capability using real sample images. Test images not used during training were provided to the system through the image upload interface.

For each input image, the system generated:

Disease status (Healthy / LSD-Affected)

Confidence score

Severity level (Healthy / Severe)

The results demonstrate the system’s ability to correctly identify LSD symptoms and provide interpretable outputs.

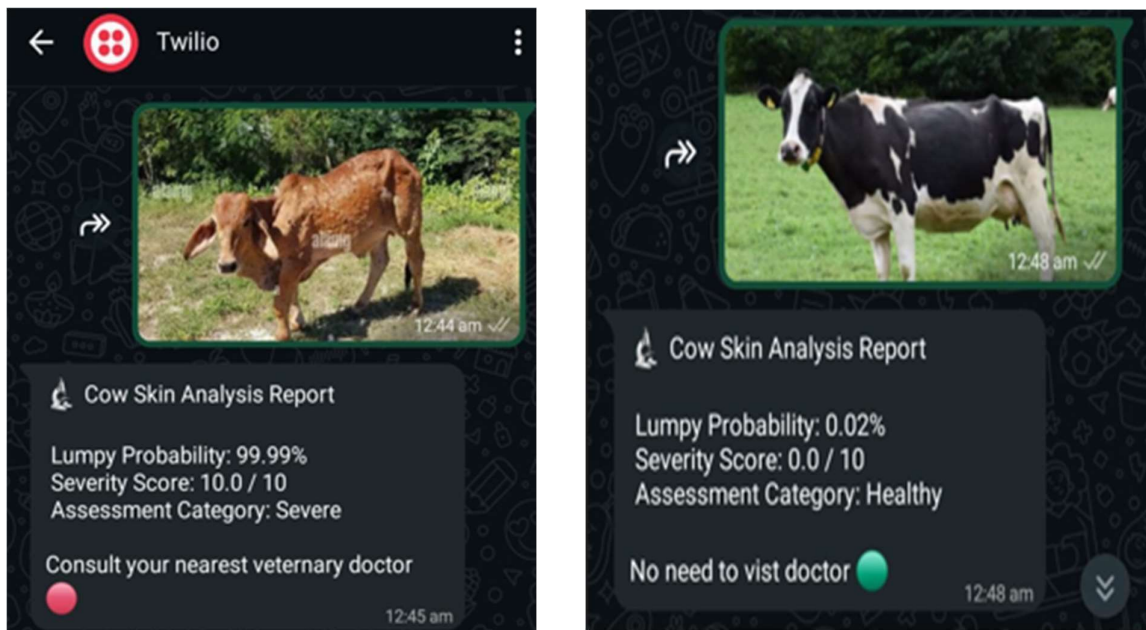


Fig 6.4: Sample Prediction Results

6.5 System Output Screenshots

This section visually demonstrates how the system interacts with users and delivers results. Screenshots help examiners understand the real-time working of the application.

The screenshots clearly show the end-to-end flow:

Image upload by farmer

WhatsApp bot interaction

Disease prediction result

Severity analysis output

Recommendation message

This proves that the system is fully functional and user-friendly.

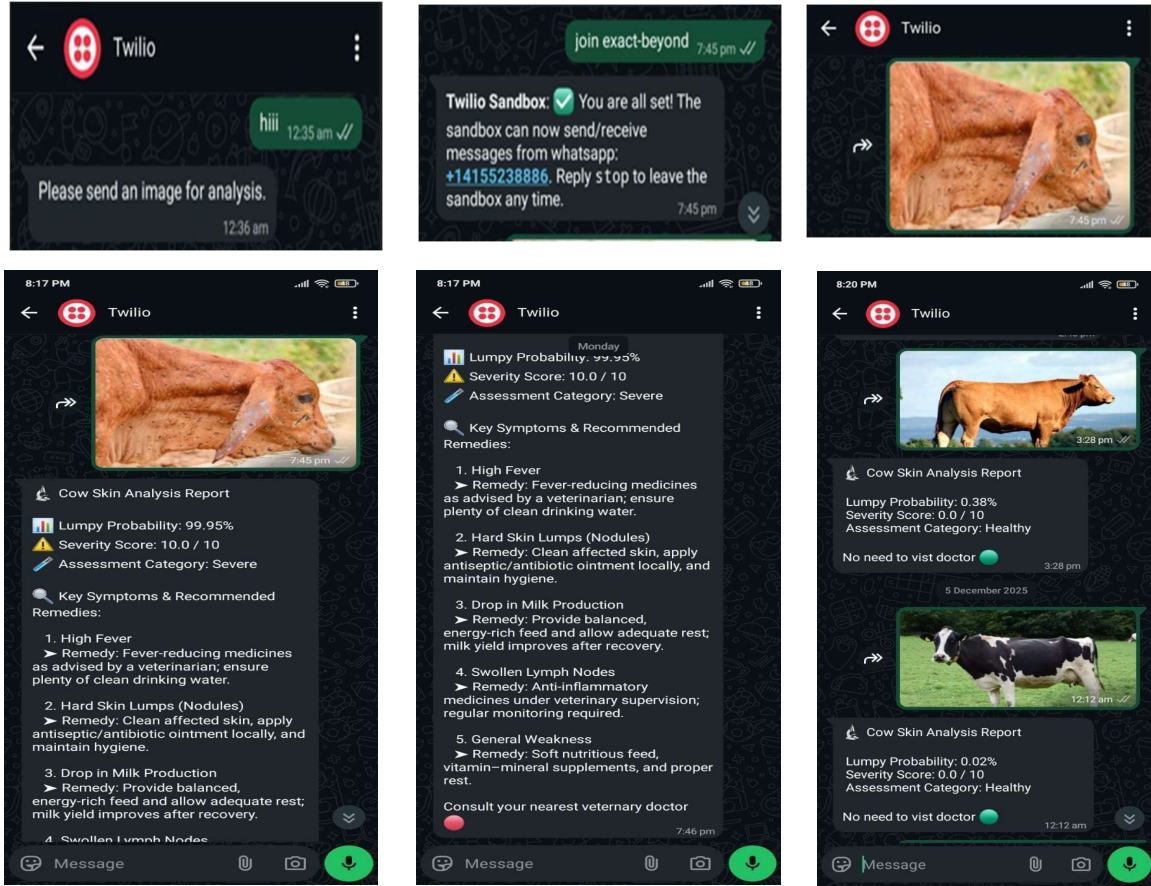


Fig 6.5: System Output Screenshots

6.6 Discussion of Results

This section interprets and analyzes the obtained results rather than presenting numerical values. The results indicate that the proposed system is effective in detecting Lumpy Skin Disease at an early stage. High accuracy and confidence scores confirm the robustness of the deep learning model. The WhatsApp-based interaction system ensures ease of use, making the solution practical for rural deployment.

Some limitations were observed when images were of poor quality or captured under extreme lighting conditions. However, overall performance remained satisfactory. The system demonstrates strong potential for real-world agricultural applications and disease monitoring.

CONCLUSION

The study "Prediction of Lumpy Skin Disease in Cows Using Deep Machine Learning" demonstrates how artificial intelligence may be applied successfully to enhance the management of livestock health. The system's primary goal was to assist farmers in early detection of Lumpy Skin Disease (LSD) in cows, allowing for prompt treatment and lowering financial losses. In the dairy industry, this strategy promotes improved animal care and illness prevention.

The method makes use of a carefully prepared dataset of photos of healthy and LSD-affected cows that were gathered from reputable sources. To enhance model performance and generalization, image preprocessing methods like scaling, normalization, and data augmentation were used. For binary classification, a deep learning model built on the CNN or Vision Transformer (ViT) architecture was employed. The system is appropriate for low- resource environments because transfer learning and fine-tuning helped achieve high accuracy while lowering training time and computational requirements.

While the Severity Analysis Module assesses the condition as mild or severe, the condition Prediction Module classifies images and assigns confidence scores. Even without technical expertise, farmers can upload photos and get results with ease thanks to a WhatsApp-based interface. The findings of the experiment show strong recall, accuracy, and precision. Future advancements could include multi-disease detection, improved mobile interfaces, and improved datasets, despite issues like internet dependence and image quality.

FUTURE WORK

Future Enhancements of the System

In addition to scalability, several enhancements can further improve system effectiveness:

1. Multi-Disease Detection

- Extend the model to detect multiple cattle diseases.
- Classify different skin infections and health conditions.
- Reduce dependency on separate diagnostic systems.

2. Integration with Veterinary Systems

- Direct alerts to nearby veterinarians.
- Appointment scheduling through the app.
- Automated treatment and vaccination recommendations.

3. AI Model Improvements

- Continuous learning using new data.
- Ensemble models for higher accuracy.
- Explainable AI (XAI) for better transparency.

4. Data Analytics and Reporting

- Historical trend analysis.
- Monthly health reports for farms.
- Predictive analytics for disease outbreaks.

The dataset was divided into training and validation sets, and unseen images were used during testing to evaluate real-time prediction capability.

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